



RIVER VALLEY HIGH SCHOOL

JC 2

PRELIMINARY EXAMINATION

CANDIDATE
NAME

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CENTRE
NUMBER

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CLASS

20J

INDEX
NUMBER

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H2 BIOLOGY

9744/02

Paper 2 Structured Questions

13 September 2021

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
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7	
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9	
10	
11	
Total	

Answer **all** questions.

- 1 Guard cells are cells surrounding each stoma found on the underside of leaves in plants. Opening of stoma is regulated by an increase in turgidity of guard cells. Fig. 1.1 shows a series of processes that occur at the cell surface membrane of a guard cell.

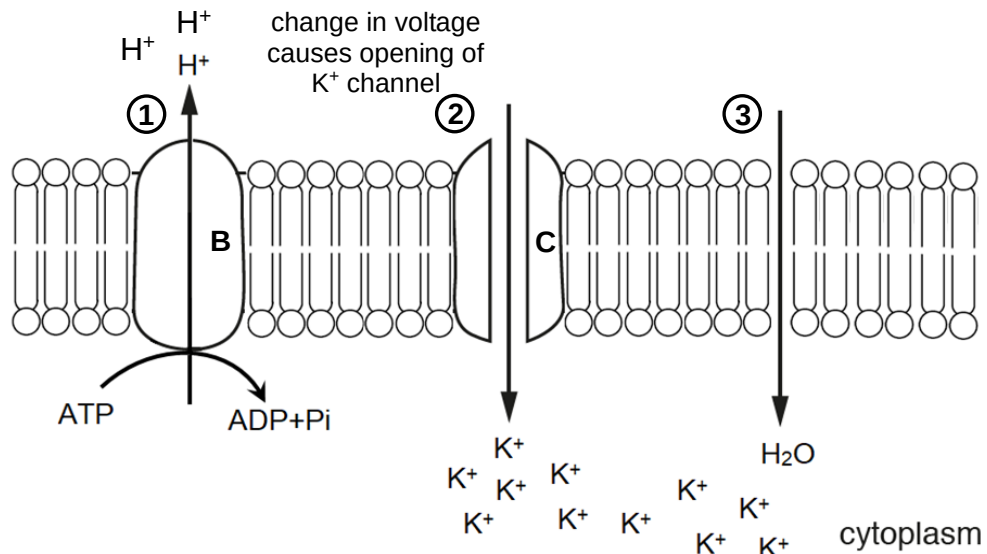


Fig. 1.1

- (a) Describe how the arrangement of phospholipids allows the maintenance of different concentrations of H^+ and K^+ across the cell surface membrane, [2]

1. Phospholipids form a bilayer;
2. The non-polar hydrocarbon chains/tail of the phospholipid molecule;
3. creates a hydrophobic core;
4. which is impermeable to ions;

AO
A

- (b) Voltage is the electrical potential difference across the membrane which is influenced by the concentration of ions.

With reference to Fig. 1.1, outline how the change in voltage results in the opening of a stoma. [3]

1. accumulation of H^+ causes K^+ channel to open;
A: H^+ causes the extracellular environment to be more positively charged
2. K^+ enters the cell;
3. by facilitated diffusion (through C);
4. water potential in cell becomes more negative / solute potential becomes more negative in cytosol;
5. water enter cells;
6. by osmosis;

AO
B

7. cell volume increases;
resulting in increase turgidity leading to opening of stoma

(c) Describe **two** structural similarities between structures of **B** and **C** that enable them to perform their functions. [2]

1. Both are transmembrane proteins, to provide a hydrophilic pathway for ions to move across the phospholipid bilayer;; AO B
2. Both have a hydrophobic portion/amino acids with non-polar R groups, to embed protein in the membrane via hydrophobic interactions;;
3. Both have a binding site, specific to a particular ions;;

Any two

A study was carried out to investigate the membrane permeability of an artificial membrane that lacked proteins.

Table 1.1 shows the results of this study and the radius of each substance tested.

Table 1.1

substance	membrane permeability / nm s ⁻¹	radius of one molecule or ion / nm
sodium ion	10 ⁻⁵	0.10
oxygen	>10 ⁵	0.12
glycerol	10 ¹	0.31
glucose	10 ⁻³	0.36
albumin protein	0	3.55

(d) Describe how properties of substances affect membrane permeability and support your answer with evidence from Table 1.1. [3]

Size:

1. substances with a radius of less than 0.36 nm / substances with small radius can pass through the membrane;
2. Evidence: albumin protein with a radius of 3.55 nm cannot move pass through the membrane;

Accept: reverse argument

Property:

3. uncharged substance can pass through the membrane compared to charged substances of similar size; AO B

4. Evidence: oxygen and sodium ions have similar radius of 0.1-0.12 nm but membrane permeability to oxygen is 10^{10} times higher than sodium ions;
5. non-polar substances can pass through the membrane compared to polar substances of similar size;
6. Evidence: glycerol and glucose have similar radius of 0.31-0.36 nm but membrane permeable to glycerol is 10^4 times higher than glucose;

[Total: 10]

AOA - 2

AOB - 8

- 2 Starch is a storage polysaccharide found in plants. Fig. 2.1 shows a representation of a starch granule.

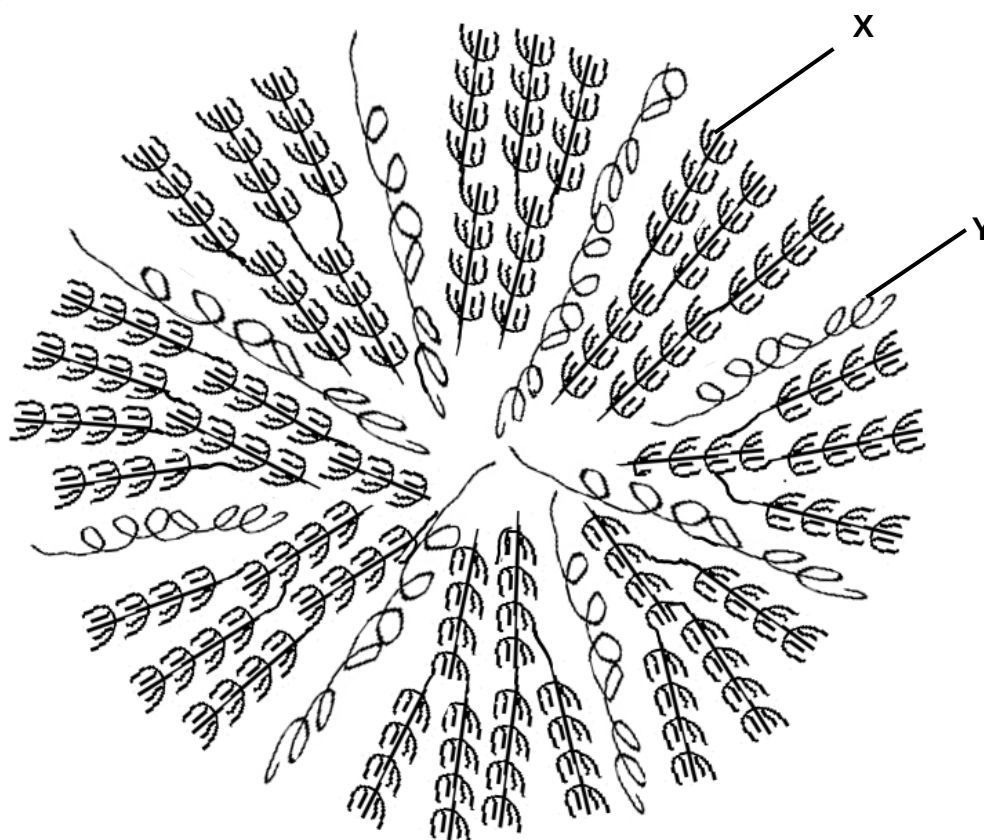


Fig. 2.1

- (a) Identify molecules X and Y.

[2]

X: amylopectin;;

AO
B

Y: amylose;;

- (b) Explain how the structural features of starch visible in Fig. 2.1 are related to its storage role in plants.

[2]

structure	property
large molecule	insoluble in water / does not exert osmotic pressure;; stores large amount of energy;;
helical coil <u>in Y</u>	compact for storage;;
<u>X</u> is highly branched	many sites for hydrolysis for rapid release of glucose molecules;; OR compact for storage;;

AO
A

Max 2

- (c) Explain how the structure of molecule **Y** differs from that of haemoglobin. [2]

Differences

AO
B

Feature	Molecule Y	Haemoglobin
Monomers;;	α -glucose	amino acid
No. of types of monomers;;	one type	20 types
Bond between monomers;;	α -1,4-glycosidic bonds	peptide bonds
No. of chains;;	one	four
Shape;;	helical	globular

Max 2

Cellulose is another polysaccharide found in plants.

- (d) In the space below, show how a glucose molecule is released from part of a cellulose chain consisting of three monomers. [3]

1. trisaccharide (3 β -glucose and correct position of bonds);;
2. correct labelling of bond;
3. water molecule added / hydrolysis;
4. free β -glucose molecule with OH group on carbon atom 1;
5. and a disaccharide;

AO
A

[Total: 9]

AOA - 5

AOB - 4

- 3 (a) State the function of DNA and describe how its properties allow it to perform this function. [3]

1. DNA stores genetic information;;
2. DNA must be chemically stable to encode information without being easily changed by age/nutrition/ environment;;
3. DNA must be able to replicate accurately so that the information can be pass down to the next generation;;
4. DNA must be capable of variation to allow for evolutionary change in a population.;;

max 2

AO
A

Fig. 3.1 shows the process by which DNA is replicated.

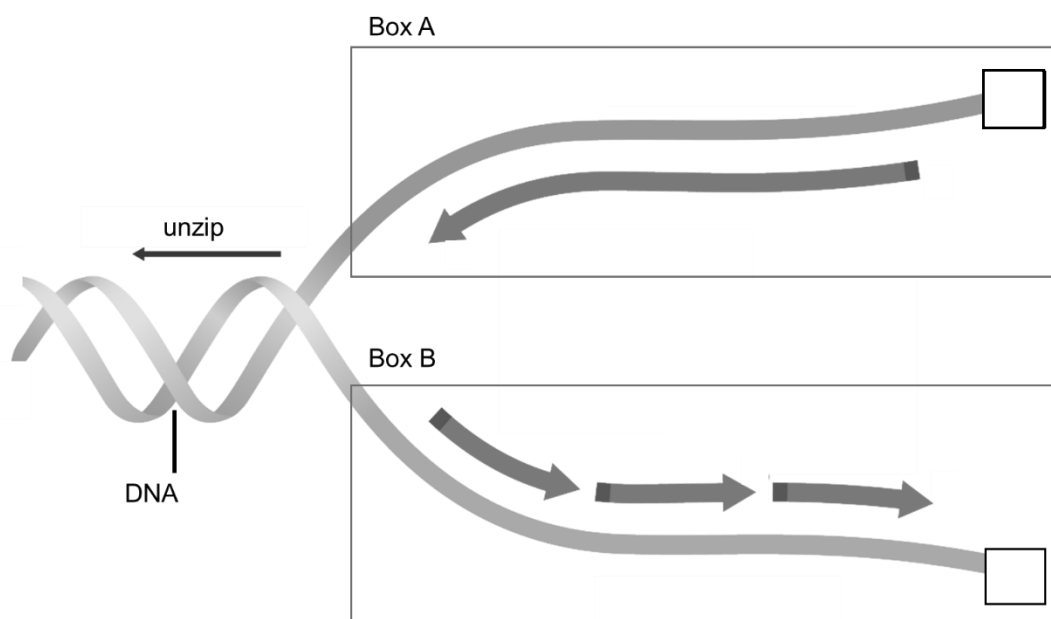


Fig. 3.1

- (b) (i) Label the 3' and 5' ends of DNA in the boxes provided in Fig. 3.1. [1]
Correct 5';
Correct 3;
- (ii) Account for the difference in how DNA is replicated in Box A and Box B. [3]

Difference

1. In box A, DNA strand is synthesised continuously but in box B, DNA strand is synthesised discontinuously/in okazaki fragments;;

Explanation

2. DNA polymerase is specific in action/specific shape of active site;
3. Only adding nucleotides to free 3'-OH end of existing strand;
4. DNA strands are antiparallel;

AO
A

5. In order to replicate DNA template in box B, a new primer is needed;
6. Each time the replication fork moves/replication bubble opens;

Antibiotics are commonly used in modern medicine to kill bacterial pathogens. Quinolones are a type of antibiotics and its mechanism of action is shown in Fig. 3.2.

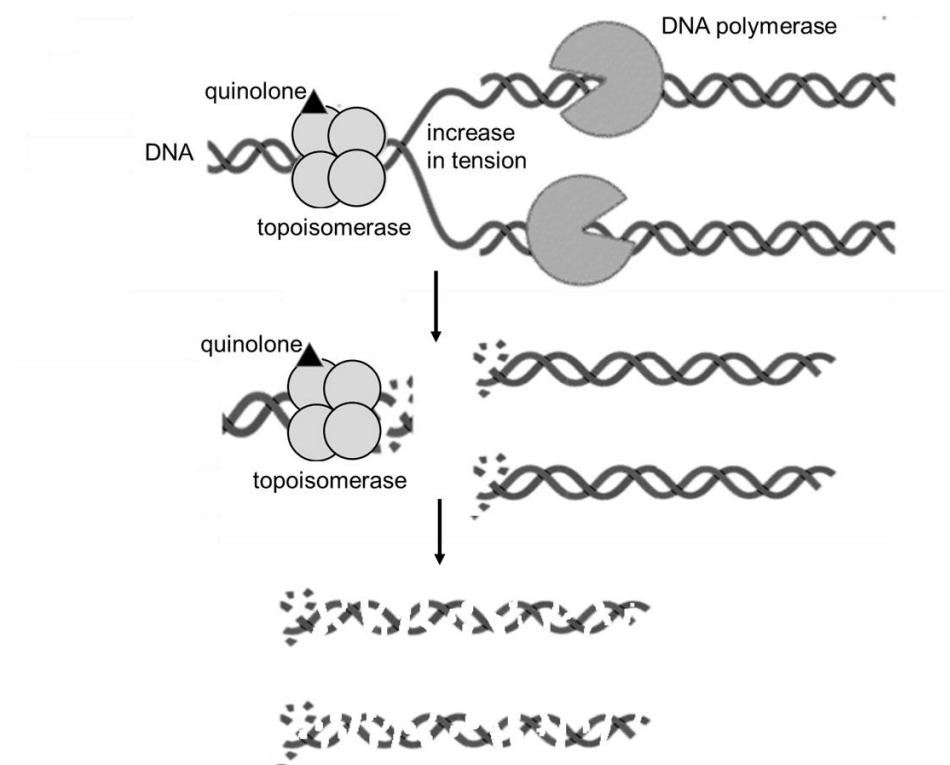


Fig. 3.2

- (c) With reference to Fig. 3.2 and your knowledge of DNA replication, describe how quinolone leads to bacterial cell death. [3]

1. Quinolone binds to topoisomerase;
2. Prevent unwinding of double helix at replication fork (of DNA molecule);
3. Increased tension;
4. Resulting in (double-strand) DNA breakage;
5. DNA molecules fragmented/broken down;
6. Genes (for survival) unable to be expressed (resulting in cell death);

AO
B

[Total: 10]

AOA - 6

AOB - 4

- 4 *BRCA1* is a tumour suppressor gene that is involved in DNA repair and transcriptional regulation. Inherited mutations in the *BRCA1* gene are mainly linked to increased breast and ovarian cancer risks. It is estimated that about 3% of breast cancers (7500 women per year) and about 10% of ovarian cancers (2000 women per year) result from inherited mutations in the *BRCA1* gene.

Table 4.1 shows the analysis of mutation types in the *BRCA1* gene in a sample of 53 individuals with heritable **breast cancer**.

Table. 4.1

mutation type	number of samples with mutation	percentage
substitution nonsense	6	11.3
substitution missense	28	52.8
insertion in frame	0	0.0
insertion frameshift	5	9.4
deletion in frame	1	1.9
deletion frameshift	7	13.2
other	6	11.3
total unique samples	53	

Table 4.2 shows the analysis of mutation types in the *BRCA1* gene in a sample of 89 individuals with heritable **ovarian cancer**.

Table. 4.2

mutation type	number of samples with mutation	percentage
substitution nonsense	24	27.0
substitution missense	29	32.6
insertion in frame	0	0.0
insertion frameshift	7	7.9
deletion in frame	0	0.0
deletion frameshift	21	23.6
other	8	9.0
total unique samples	89	

Adapted from: Liang *et al*, International Journal of Molecular Medicine (2018)

- (a) (i) Calculate the total number of individuals per year who develop **breast** and **ovarian** cancers due to substitution mutations.

Show your working clearly.

[2]

Calculate total number who develop breast cancer:

- Correct formula;

Calculate total number who develop ovarian cancer:

- Correct formula;

Provides correct total number:

- Correct answer & precision;;

Total number who develop breast cancer

$$= (6+28)/53 \times 7500 \quad \text{or use given \%} = 4807$$

$$= 4811.32$$

$$= 4811 \text{ (whole no.)}$$

Total number who develop ovarian cancer

$$= (24+29)/89 \times 2000 \quad \text{or use given \%} = 1192$$

$$= 1191.01$$

$$= 1191 \text{ (whole no.)}$$

Total number of individuals per year who develop breast and ovarian cancers due to substitution mutations

$$= 4811 + 1191 \quad \text{or use given \%} = 5999$$

$$= 6002$$

AO
B

- (ii) The deletion in frame of *BRCA1* gene in the sample for **breast** cancer resulted in a BRCA1 protein that is 23 amino acids longer.

Explain how this occurs.

[2]

1. Deletion of splice site;;

2. Intron is not removed during splicing / intron joined with exons;

3. intron translated (with coding sequence);

OR

4. Deletion of stop codon;;

5. Translation continues;

6. until ribosome reaches downstream stop codon;

AO
B

Mutation screening, using gel electrophoresis, was performed to compare the nucleotide sequences of a normal and a mutated *BRCA1* gene.

Fig. 4.1 shows part of the sequence from the gel electrophoresis.

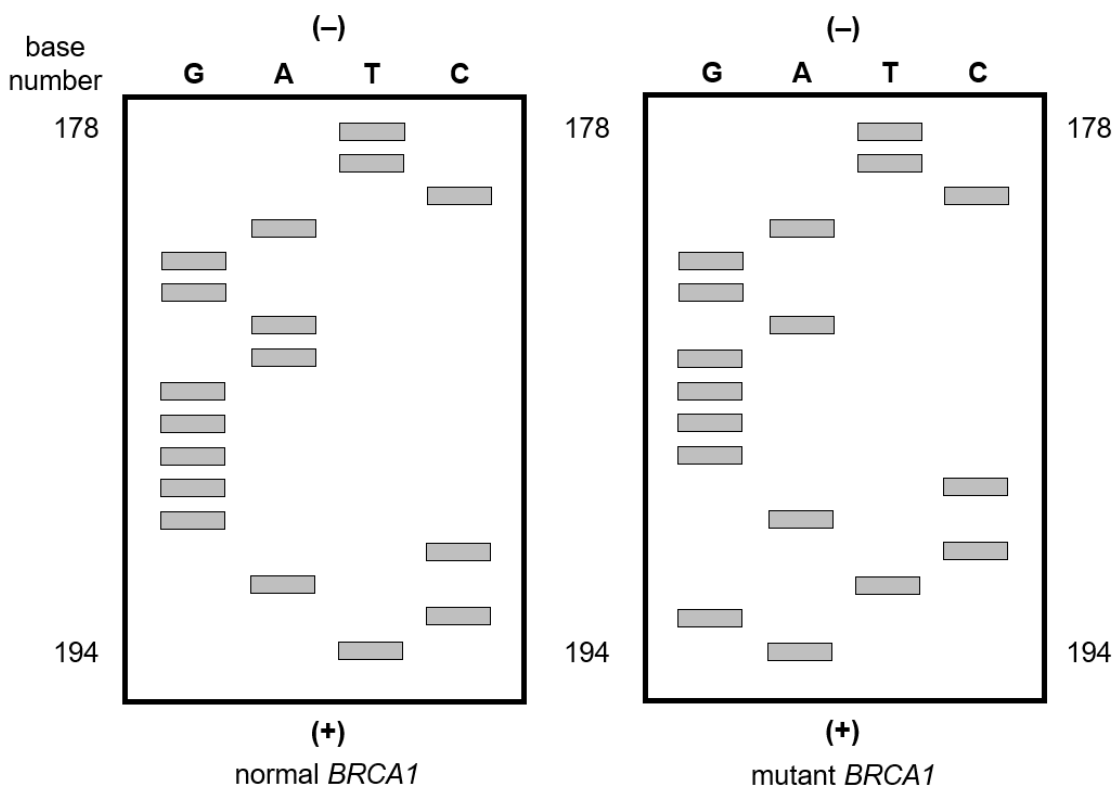


Fig. 4.1

- (b) Using the information provided in Fig. 4.1, describe precisely the mutation and explain how it results in the formation of cancer. [4]

1. Deletion of 2 nucleotides / A & G;
2. At nucleotide positions 184/185 and 186/187/188/189/190;
3. Resulting in frameshift mutation;
4. Formation of premature stop codon;
5. Truncated *BRCA1* polypeptide / non-functional *BRCA1* protein;
6. No DNA repair;
7. No transcription of genes coding for cell cycle arrest/apoptosis;

OR

8. Loss-of-function mutation;
9. Of both alleles of the tumour suppressor gene;

10. Leading to uncontrolled cell division;

AO
B

(c) Other than tumor suppressor genes, oncogenes have a key role in cancer.

Explain what is meant by an oncogene in the context of cancer formation. [2]

1. Produced by gain-of-function mutation of proto-oncogene;
2. Codes for hyperactive / degradation-resistant protein;
3. Or excess of normal proteins;
4. Resulting in overstimulation of cell division;

AO
A

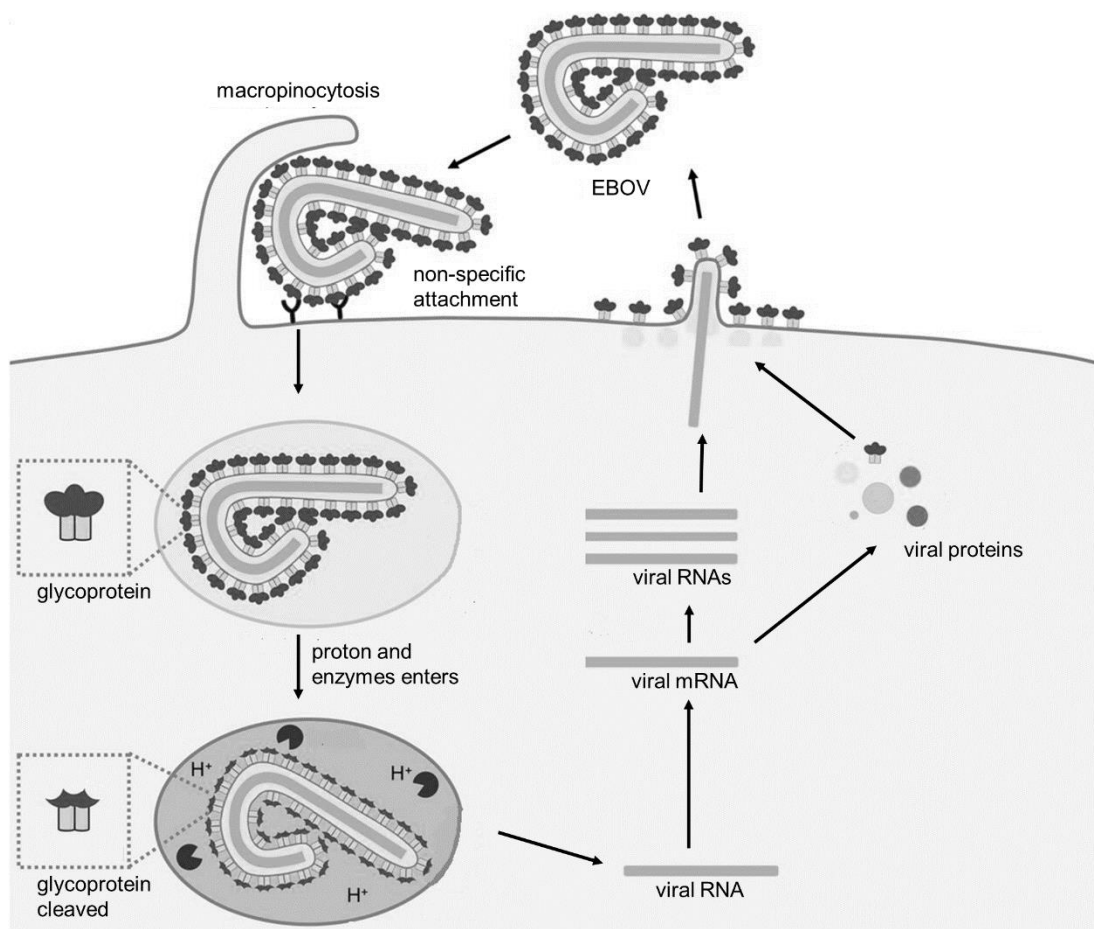
[Total: 10]

AOA – 2

AOB – 8

- 5 Ebola is a rare yet deadly infectious disease. The virus that causes Ebola is known as the Ebola virus (EBOV). EBOV is an enveloped virus containing a cylindrical nucleocapsid that encloses a RNA genome and a viral enzyme.

Fig. 5.1 shows part of the reproductive cycle of EBOV.



Adapted from Sullivan *et al*, The Journal of Immunology, 201 (3) 833-842, August 1 (2018)

Fig. 5.1

- (a) Explain why viruses, such as EBOV, are obligate parasites. [1]

Requires a host cell in order to reproduce/unable to reproduce outside of a host cell;; AO
A

- (b) With reference to Fig. 5.1 and your knowledge of HIV,
(i) distinguish between the entry of EBOV to that of HIV into its host cell, [3]

Differences

Stages	EBOV	HIV
Entry	Via macropinocytosis	Enters the cell via fusion of hostcell surface membrane and viral envelope;;

AO
B

Fate of viral envelope proteins	GP on viral envelope is cleaved/ GP enters the host cell	Gp120 & gp41 are not cleaved/ Gp120 & gp41 remains on host cell surface membrane;;
Attachment	GP binds to receptors on host cell surface membrane non-specifically, not requiring complementary shape	HIV glycoprotein 120 binds to (CD4) receptors on host cell surface membrane via complementary shape;;

- (ii) suggest the identity of the enzyme present in the nucleocapsid of EBOV and explain why this enzyme is needed to form new virions. [3]

1. RNA-dependent RNA polymerase;;
2. The host cell does not contain RNA polymerase that can use RNA as a template;
3. to transcribe the viral RNA into viral mRNA;
4. that host cell ribosome uses for translation;
5. and replication / make more copies of the viral RNA;

AO
B

- (c) EBOV acquires its viral envelope using the same process as HIV.

Describe how EBOV acquires its viral envelope.

[3]

1. glycoproteins and other viral proteins are synthesized and;
2. transported to host cell surface membrane (in vesicles);
3. to form 'exit points' for new viral particles;
4. host cell surface membrane surrounds viral particles;
5. immature viral particles bud off from host cell at 'exit points';
6. thus acquires viral envelope with glycoprotein from host cell surface membrane;

AO
A

[Total: 10]

AOA – 4

AOB - 6

- 6 The formation of the *lac* operon was an important adaptation of prokaryotes in response to its environment. This organisation of genome contributed to an increased evolutionary advantage for prokaryotes.

(a) Describe how the *lac* operon contributed to increased evolutionary advantage. [4]

1. Grouping of *lac Z*, *lac Y* and *lac A*;
2. Which are involved in catabolic pathway of lactose;
3. Into a single transcription unit / under same promoter;
4. Allows regulation of expression of structural genes at the same time;
5. Therefore increasing speed/efficiency of response;
6. Allows expression of structural genes when low glucose;
7. and high lactose;
8. Therefore conserving/preventing wastage of energy and resources;
9. Higher chance of survival and reproduction;

AO

A –

2

AO

B –

2

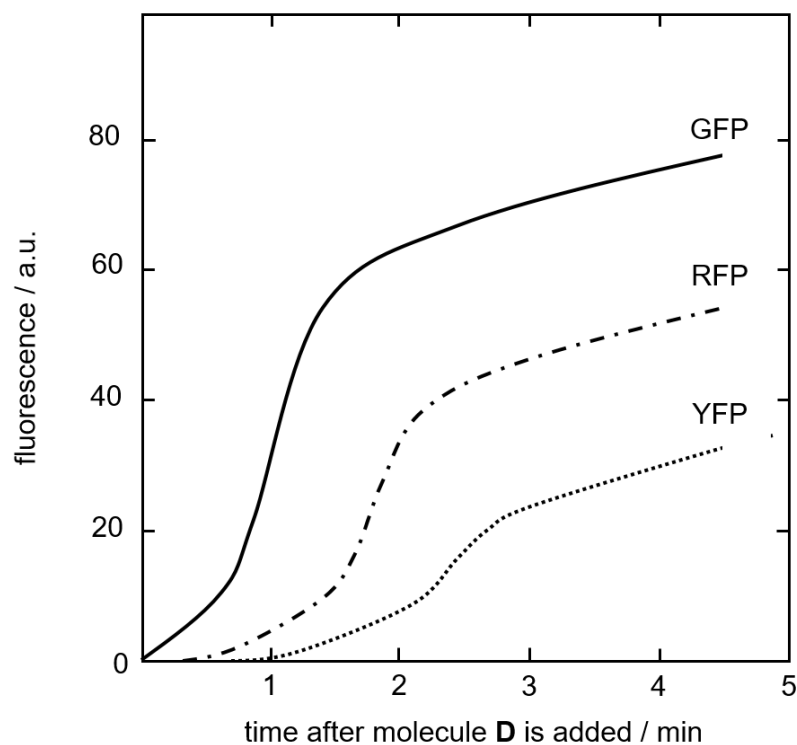
In an investigation on the effect of gene position and rate of expression, the structural genes in the *lac* operon were replaced by genes coding for green fluorescence protein (GFP), red fluorescence protein (RFP) and yellow fluorescence protein (YFP). This allowed for the observation of expression of structural genes in the *lac* operon.

Fig. 6.1 shows the structure of the modified *lac* operon.



Fig. 6.1

The modified *lac* operon was inserted into a prokaryotic cell. This prokaryotic cell has no genome and contains only molecular machinery for gene expression and an excess of catabolite activator protein and lac repressor protein. The operon was induced to turn on by adding molecule **D**. Fig. 6.2 shows the results of the experiment.



Adapted from: Hussein *et al*, PNAS (2011)

Fig. 6.2

- (b) Describe how molecule **D** induces the synthesis of GFP protein. [3]

Molecule **D** is allolactose

1. Which binds to allosteric site on lac repressor;
2. Resulting in change to inactive conformation;
3. Lac repressor cannot bind to operator of *lac* operon;
4. RNA polymerase able to bind promoter;
5. To transcribe *GFP* gene;
6. And translate mRNA to form GFP protein;

AO
A

- (c) (i) Explain the differences in GFP and YFP fluorescence. [2]

Differences

1. GFP is expressed before YFP;
2. GFP is expressed initially at a higher rate than YFP;
3. Total amount of GFP expressed is higher than that of YFP;

Explain

4. Because transcription of genes from GFP to YFP;
5. GFP mRNA is translated first;
6. And for a longer duration; □

Max. 2

AO
B

- (ii) Suggest the significance of this finding on GFP in (c)(i) to the uptake of lactose by prokaryotes. [1]

More β -galactosidase produced, high rates of breakdown/conversion of lactose to maintain gradient for entry of lactose into cell;; AO
B

[Total: 10]

AOA – 5

AOB - 5

- 7 In domestic pigeons, the presence of feathers on their head is known as a crest. This is determined by a gene found on an autosomal chromosome.

In domestic pigeons, the sex chromosomes are Z and W, and the heterogametic sex is the female. Thus, the female has one Z and one W chromosome (ZW) while the male has two Z chromosomes (ZZ). The gene controlling plumage colour of the pigeon is found on chromosome Z.

Table 7.1 shows the phenotypes of 13 offsprings from the cross of an ash-red female pigeon with no crest and an ash-red male pigeon with crest.

Table 7.1

offspring number	sex	colour	presence of crest
1	male	ash-red	no crest
2	male	ash-red	no crest
3	male	ash-red	no crest
4	male	ash-red	no crest
5	male	ash-red	no crest
6	male	ash-red	no crest
7	male	ash-red	no crest
8	female	ash-red	no crest
9	female	ash-red	no crest
10	female	blue	no crest
11	female	blue	no crest
12	female	blue	no crest
13	female	blue	no crest

- (a) State the dominant allele for both the gene loci controlling presence of crest and plumage colour. [2]

Allele for no crest;;

Allele for ash-red;;

AO
B

- (b) Using the following symbols to represent the alleles, draw a genetic diagram to explain the results of the cross.

Key

A – ash-red plumage

B – blue plumage

[4]

D – no crest**d** – crest

1. Parents genotype (showing sex linkage);;
2. Parents gametes;;
3. Punnett sq showing correct genotype
4. Matching phenotype;;

Parental phenotypes ash-red female, no crest ash-red male, crest

Parental genotypes $Z^A W D D$ $Z^A Z^B d d$

Gametes $W D$ $Z^A D$ $Z^A d$ $Z^B d$

Punnett square

		Male gametes	
		$Z^A d$	$Z^B d$
Female gametes	$W D$	$Z^A W D d$	$Z^B W D d$
	$Z^A D$	$Z^A Z^A D d$	$Z^A Z^B D d$

1 ash-red female, no crest : 1 blue female, no crest : 2 ash-red male, no crest

Observed numbers

2

4

7

AO
B

χ^2 value for the cross can be calculated using the following below.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where Σ = sum of

O = observed value

E = expected value

- (c) (i) Using the genetic diagram in (b) and Table 7.1, calculate the χ^2 value for the cross. [2]

Working;;

Answer;;

Critical values of this χ^2 distribution are shown in Table 7.2.

Table 7.2

	probability, p
--	----------------

degrees of freedom	0.95	0.80	0.50	0.20	0.10	0.05	0.01
1	0.004	0.06	0.46	1.64	2.71	3.84	6.64
2	0.10	0.45	1.39	3.22	4.61	5.99	9.21
3	0.35	1.01	2.37	4.64	6.25	7.82	11.35
4	0.71	1.65	3.36	5.59	7.78	9.49	13.28

(ii) Using the calculated χ^2 value in (c)(i) and Table 7.2, explain what conclusion can be drawn from the results.

[2]

1. Given calculated $\chi^2 = 0.692$, $v = 2$, the probability of the deviation is between 0.50 and 0.80;
2. which is greater than 0.05;
3. do not reject H_0 ;
4. There is no significant difference between the observed and expected results;

AO
B

[Total: 10]

AOA - 0

AOB - 10

- 8 Fig 8.1 shows part of the process of aerobic respiration in *S. aureus*, a gram positive bacteria.

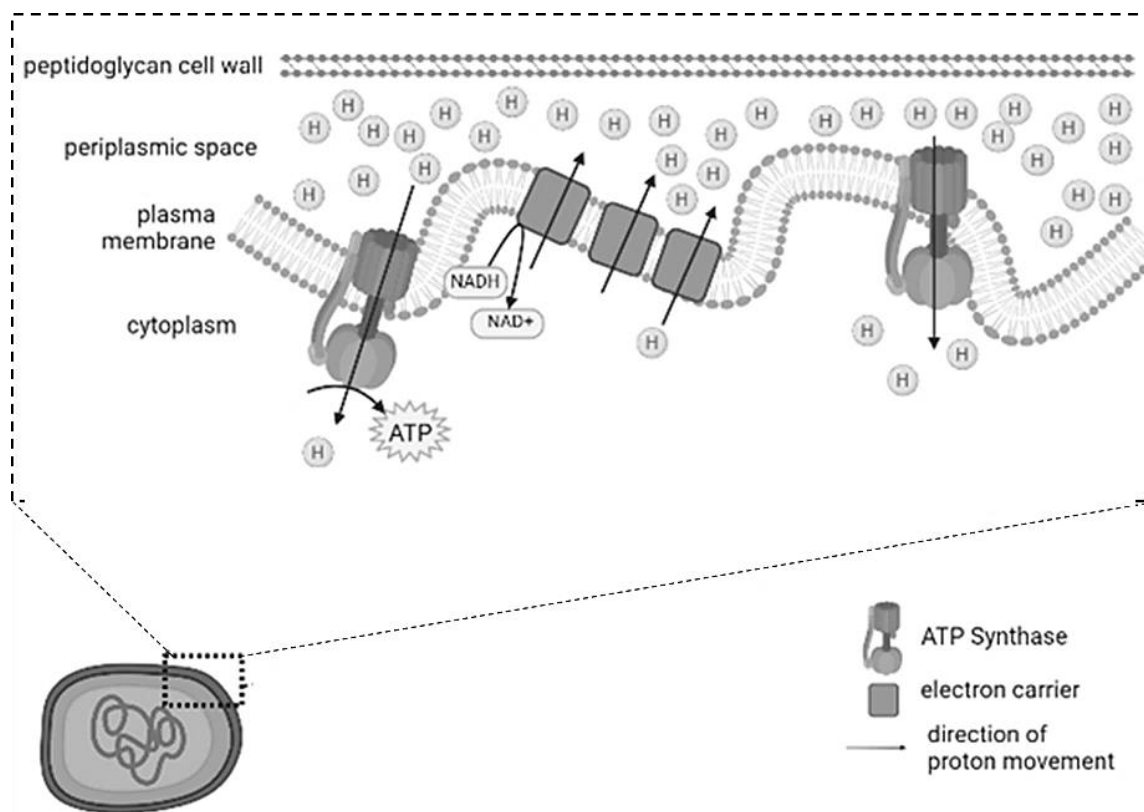


Fig 8.1

- (a) Glucose enters bacteria cells via glucose transporters found on the plasma membrane. Glucose then undergoes glycolysis.

Outline briefly the process of glycolysis for one molecule of glucose.

[3]

1. Glucose undergoes phosphorylation/activated by ATP;
2. and is oxidised;
3. to give 2 molecules of NADH;
4. and substrate level phosphorylation;
5. to give 4 ATP/net of 2 ATP per glucose;
6. and two molecules of pyruvate;

AO
A

- (b) Distinguish between the process shown in Fig 8.1 and non-cyclic photophosphorylation in plant cells.

[4]

Differences;;

	ATP synthesis in bacteria	Non-cyclic photophosphorylation
Electron source	Respiratory substrates	Water;;
Photosystem	Does not involve photosystems	Involves photosystems;;

A
O
B

Electron & proton carrier used	NADH	NADP+;;
Reaction in which electron and proton carrier is involved	NADH oxidized to NAD	NADP+ reduced to NADPH;;
Energy source	Oxidation of respiratory substrates	Light energy;;
Direction of proton flow down ATP synthase	Periplasmic space to cytoplasm	Thylakoid lumen to stroma;;
Direction in which protons are pumped	Cytoplasm to periplasmic space	Stroma to thylakoid lumen;;

(c) Describe the role of NAD in aerobic respiration.

[2]

1. Act as co-enzyme of dehydrogenase / electron and proton carrier;;
2. Accepts the electrons and protons (hydrogen atoms) from respiratory substrates;
3. In glycolysis, link reaction and Krebs cycle;
4. Passes high energy electrons to electron carriers in the Electron Transport Chain (ETC);

AO
A

[Total: 9]

AOA – 5

AOB - 4

9 (a) Describe how micro-evolution is related to macro-evolution. [2]

1. changes in allele frequency in gene pool of a population over generations (micro-evolution);
2. may lead to reproductive isolating mechanisms forming between populations;
3. preventing interbreeding / gene flow;
4. leading to speciation / forming of new species;

AO
A

Modern varieties of wheat have developed from numerous hybridisation events between different species of wild grasses. Fig. 9.1 shows some of the possible steps that are believed to have been involved in the development of bread wheat, *Triticum aestivum*.

The letters **A**, **B** and **C** represent three different sets of seven chromosomes. For example, in *Triticum urartu* has two **A** and therefore possesses fourteen chromosomes.

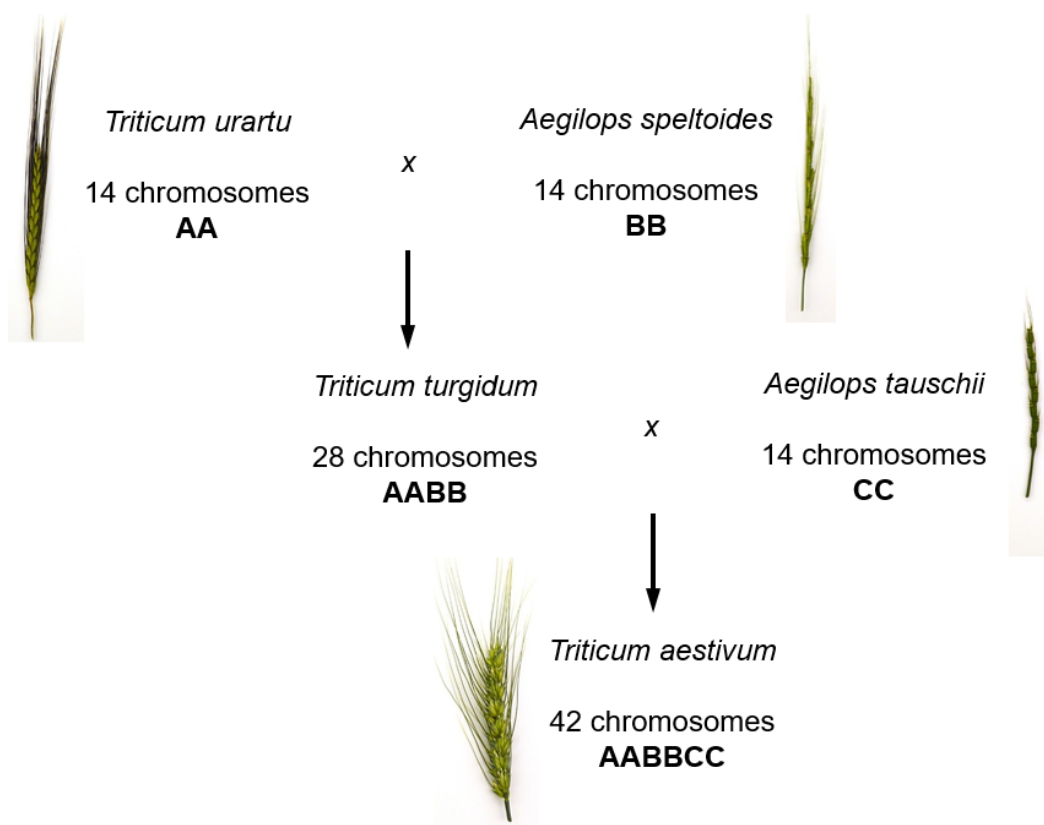


Fig. 9.1

(b) With reference to Fig. 9.1, explain how the new species of bread wheat, *Triticum aestivum*, formed from *Triticum turgidum*. [4]

1. Hybridisation between *Triticum turgidum* and *Aegilops tauschii*;
2. Gamete containing 14 chromosomes / **AB** (from *Triticum turgidum*);
3. Fuses with gamete containing 7 chromosomes / **C** (from *Aegilops tauschii*);
4. Forming hybrid zygote containing 21 chromosomes / **ABC**;
5. Doubling of chromosomes;

AO
B

6. To form 42 chromosomes / **AABBCC**;
7. Allopolyploid / physiological isolation;
8. Unable to interbreed with parents; (SLK: Accept genetic species concept?)

- (c) **A, B and C** contain common genes that code for the same proteins essential to the survival of modern wheat. Recessive alleles of these genes were found to be more prevalent in the population of *Triticum aestivum* compared to the other varieties.

Explain why recessive alleles present in **A** may be better preserved in the population of *Triticum aestivum* than in that of *Triticum urartu*,

[2]

1. More sets of chromosomes / presence of **B** and **C**;
2. Higher chance; of heterozygote protection;
3. By similar genes / genes coding for same characteristic;

AO
B

- (d) Explain why genetic variation is important to the survival of a species.

[2]

1. Genetic variation is an important source of raw material for evolution;
2. Increases ability of a species to survive in a changing environment;
3. Resultant phenotypic variation selected for by different selection pressures;
4. Also may allow species to inhabit a wide range of habitats;

AO
A

[Total: 10]

AOA – 4

AOB - 6

- 10 Contact tracing is a key process to slow down the spread of COVID-19.

Antigen test and antibody test detect the presence of COVID-19 virus through the presence of the COVID-19 antigen or COVID-19 antibody respectively in blood samples of individuals who are suspected to be infected with the virus. These tests are done in the effort to map the transmission network of the virus amongst individuals so that individuals who are capable of transmitting the virus can be quickly quarantined.

- (a) Describe how an antigen presenting cell presenting the COVID-19 antigen led to the production of COVID-19 antibodies.

[4]

1. Binding to (CD4) receptor on helper T lymphocytes;
2. Triggers the secretion of interleukin-1 (IL-1) by APC;
3. activating helper T lymphocytes;

AO
A

4. A naïve B lymphocyte with B Cell Receptor presents COVID19 fragment on cell surface membrane; (SLK: To include on class I MHC?)
5. An activated helper T lymphocyte binds to this B lymphocyte;
6. Releasing interleukin 2;
7. Resulting in differentiation of B lymphocytes into plasma cells;
8. plasma cells secrete COVID19 antibody molecules;

(b) Suggest why both antigen test and antibody test are necessary in the process of contact tracing to map the disease transmission network. [3]

1. An individual who has a past infection and recovered from COVID-19 may not have the COVID19 antigen but will have the COVID-19 antibody/OWTTE;;
2. Thus will be negative for antigen test but positive for the antibody test/OWTTE;;
3. However, the individual was capable of transmitting the virus before he recovered and so vital in mapping the disease transmission network amongst individual/OWTTE;;

Marking for 3 main ideas

- Difference between antibody vs antigen;;
- The test results for individuals;;
- Its importance in tracing susceptible individuals;;

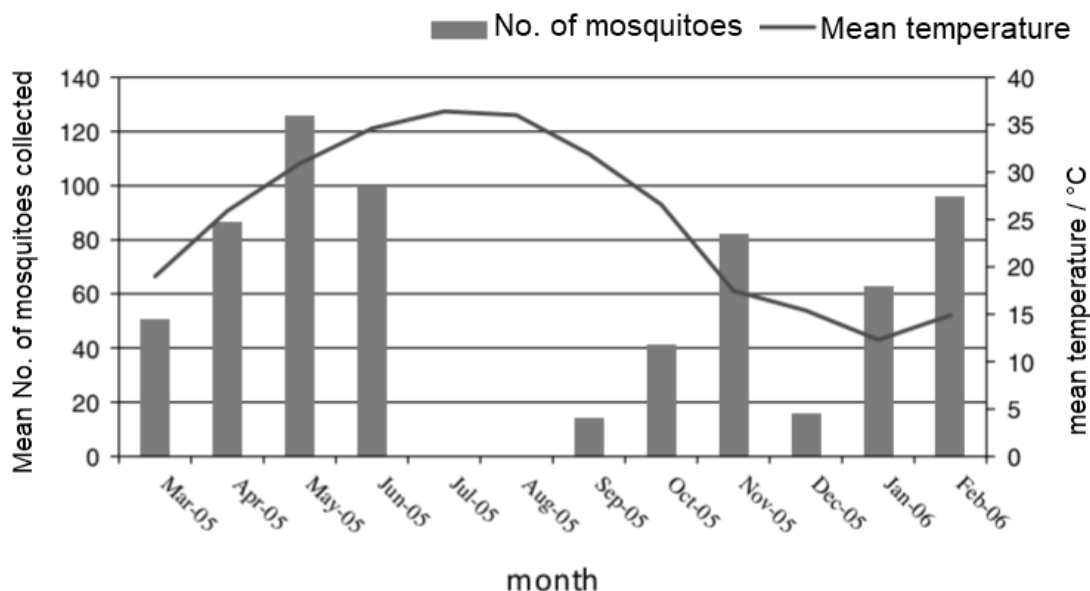
AO
B

[Total: 7]

AOA – 4

AOB - 3

- 11** Insects are the most dominant group of organisms on the planet in terms of species richness, abundance and biomass. Global warming has a marked influence on the physiology of insects. One such insect is the mosquito. Fig. 11.1 shows the effect of temperature on seasonal abundance of mosquitoes in the tropics.



Source: Mosquito fauna (Diptera: Culicidae) of the Eastern Region of Saudi Arabia and their seasonal abundance, *Journal of King Saud University - Science* · January 2010

Fig. 11.1

(a) With reference to Fig. 11.1,

(i) explain the effect of temperature on the mean number of mosquitoes between March and May 2005. [2]

1. As temperature increases, mean number of mosquitoes increases;
2. Increase metabolic rate / rate of enzyme-catalysed reactions;
3. shorten life cycle;;

AO
B

(ii) identify the maximum temperature at which the mosquitoes can survive. [1]

35°C;;

AO
B

(iii) explain why global warming can lead to spread of dengue beyond the tropics. [2]

1. Insects have narrow temperature tolerance / temperature sensitive;
A: Ref to insects cannot survive at high temperature
2. Warmer winter temperature in temperate region;
R: temperate region is cooler
3. Mosquitoes move polewards / higher latitude;

AO
A

4. increasing the number of disease vector for dengue beyond the tropics;

[Total: 5]